

Trim & Flight Model

Trimming — teaching the glider's control model to match how the vehicle actually flies — is where most of a Seaglider pilot's early-mission effort goes. Pitch trim is arguably the **single biggest influence on flight**: a well-trimmed vehicle flies cleanly on both profiles, dives and climbs in about the same time, and wastes far less energy on the VBD; a badly trimmed one burns battery, and sometimes doesn't fly at all. This page covers the practical trimming workflow (pitch, then buoyancy, then roll), pre-deployment ballasting, and the basestation's automated Flight Model System (FMS).

Source

Paraphrased from the APL-UW IOP office-hours series (pitch trimming; ballasting and volmax; flight-model questions) and webinar series (*Piloting with basestation3*), the UW *Determining Seaglider Velocities Automatically* flight-model paper (Bennett, Stahr & Eriksen), and the APL-UW *SGX Documentation*. Numbers quoted here are IOP starting points, not gospel — confirm parameter definitions in the [Parameter Reference Manual](#) and defer to APL-UW IOP guidance.

Ballasting before the mission: estimating volmax in a tank

Before a sea trial, the glider needs lead trimmed so that its maximum volume (`volmax`) gives the right thrust in the density of the operating area. The full tank procedure — weighing in air and water, computing `volmax` at several VBD positions, and converting it into a lead adjustment — now lives on its own page: see [Seaglider Ballasting Procedure](#).

The one constant from that procedure worth carrying over here, since the rest of this page leans on it repeatedly: `VBD_CNV = -0.2453 cc/AD count`, with `$VBD_MIN` (~400 counts) as bladder **full** and `$VBD_MAX` (~3960 counts) as bladder **empty** — smaller counts mean *more* volume, not less.